
Recursive Identification of Structured Systems: An Instrumental-Variable Approach applied to Mechanical Systems¹

Abstract: Many systems, including mechanical systems, rely on parametric models for analysis, advanced control, and online monitoring, particularly as they experience temporal changes due to varying operating conditions, degradation, external disturbances, or dynamic interactions, necessitating the monitoring of their time-varying dynamics through online estimation. The objective of this chapter is to develop a real-time estimation algorithm aimed at estimating time-varying dynamics within an interpretable model structure. To achieve this, an additive parameterization is adopted, offering enhanced parsimony, particularly suitable for mechanical systems. The proposed approach integrates the recursive simplified refined instrumental variable method with block-coordinate descent to minimize an exponentially-weighted output error cost function. This novel recursive identification method yields parametric continuous-time additive models applicable in both open-loop and closed-loop controlled systems. Numerical simulations demonstrate the effectiveness of the proposed method, further validated using experimental data to detect time-varying resonance mechanics.

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8.1 Introduction

System identification involves obtaining mathematical models of complex phenomena from measured data, with wide-ranging applications across diverse fields [161, 236]. The development of accurate models enables the simulation, analysis, and prediction of complex system behavior. In control engineering, the models are key in designing effective control strategies, while its predictive capabilities support decision-making and planning.

Online system identification plays a significant role in dynamic environments where systems are prone to change over time [159, 277]. Unlike offline identification, which relies on a batch of data, online system identification continuously updates models in real-time as new data becomes available. Online system identification allows for quick adjustments to evolving conditions, which can enhance model accuracy, improve performance, and facilitate effective maintenance planning of various industrial process [40]. Consequently, it is instrumental in maintaining optimal system performance, enabling timely fault detection and diagnosis, and supporting adaptive control in dynamic unpredictable settings.

Estimation of continuous-time (CT) dynamical models of industrial processes offers advantages over discrete-time (DT) models. Continuous-time models can provide a more accurate representation of the underlying dynamics of physical systems, since they allow for the direct incorporation of *a priori* knowledge such as the relative degree of the physical systems they model [89]. The incorporation of prior knowledge in the form of interpretable physics has gained increasing importance in the fields of system identification, control, monitoring, and machine learning. In the context of linear system identification, incorporating relative degree information is particularly useful when the time evolution of the signals is naturally continuous, as seen in mechanical systems [91], where impulse response discontinuities are typically absent due to the double integration relationship between force and position.

While a single non-additive transfer function is commonly employed in linear system identification, practical applications in flexible motion systems [184, 268], and vibration analysis are more effectively conceptualized as a sum of transfer functions with distinct denominators. For example, mechanical systems are more naturally interpreted in modal form, where each submodel represents a different resonant mode [91]. Additive model parametrizations bring benefits such as enhanced physical insight for fault diagnosis [39] and improved numerical conditioning for high-order systems [112]. These parametrizations are also explored in statistics [121] and econometrics [119, 185], offering increased model flexibility and the ability to decentralize analysis for optimization and control purposes. Despite contributions in nonlinear discrete-time finite-impulse response and generalized Hammerstein model estimation [10, 11], the utilization of additive model structures in system identification has been limited.

The identification of additive models has recently gained increasing atten-

tion. An algorithm for offline estimation of continuous-time additive models in single-input single-output (SISO) systems, applicable to both open and closed-loop configurations, has been introduced [112]. This approach is based on the simplified refined instrumental variable method, SRIVC [281], which consists of iteratively computing instrumental variable estimates of the system parameters. Alternatively, a block coordinate algorithm is utilized for identifying these systems under this structure with a tailored block-coordinate descent version of the SRIVC method [108].

In contrast to these offline algorithms, many recursive algorithms have been developed by parameterizing the model as a single non-additive transfer function [159, 277]. These online methods are typically based on a forgetting factor, or in state estimation mechanisms analogous to the Kalman filter. Most methods are categorized as modifications of offline identification methods, (non-)linear filtering methods, stochastic approximation methods, and pseudolinear regression-based methods [159]. Methods for time-varying systems, such as [189, 190] explore open and closed-loop approaches based on prefiltering the input and output derivatives with ad-hoc prefilters, similar to the state-variable filter and refined instrumental variable methods for linear and time-invariant systems [88, 281].

Although current recursive methods can effectively track time-varying systems, these yield non-additive transfer function models with limited interpretability, as the estimated parameters may not directly correspond to physical system quantities. In this chapter, additive linear time-varying systems are considered for which the parameter variation is slow compared to the system dynamics. To track these systems, instrumental variable iterations are performed for each submodel until convergence, aligning with the philosophy of the block coordinate descent approach for non-convex optimization [163]. A comprehensive identification method for modeling additive linear time-varying continuous-time systems is proposed for both open and closed-loop settings. The method is referred to as recursive SRIVC for additive systems, or abbreviated as additive RSRIVC.

In summary, the main contributions are summarized as follows.

- C1: A condition is provided under which it is preferable to estimate an additive model structure instead of an unfactored transfer function.
- C2: A recursive continuous-time system identification algorithm is derived for systems in additive form, for both open and closed-loop systems and its consistency is assessed.
- C3: The proposed method is evaluated through simulations and is experimentally validated on a flexible beam setup with a time-varying resonance mode.

The remainder of this chapter is organized as follows. In Section 8.2 the

system setup is presented and the problem is formulated. Section 8.3 elaborates on the benefits of the proposed approach, showcasing motivational examples and demonstrating where an additive decomposition is preferable to an unfactored transfer function. Section 8.4 contains the main contribution of this work, namely the derivation of a recursive identification method for continuous-time systems in additive form. The estimator is numerically evaluated in Section 8.5, and implemented on a true mechanical system in Section 8.6. Lastly, Section 8.7 concludes this chapter.

8.2 System setup and problem formulation

Initially, the system setup for additive systems in open-loop and closed-loop configuration is outlined, followed by the formulation of the estimation problem.

8.2.1 System setup

Consider the following continuous time (CT), linear and time-variant (LTV), single-input single-output (SISO) continuous-time system

$$x(t) = G^*(p, t)u(t), \quad (8.1)$$

where $G^*(p, t) = B^*(p, t)/A^*(p, t)$, with p being the Heaviside operator (i.e., $px(t) = \frac{dx(t)}{dt}$). The terms $A^*(p, t)$ and $B^*(p, t)$ are polynomials in p with time-varying coefficients of the form

$$A^*(p, t) = a_n^*(t)p^n + a_{n-1}^*(t)p^{n-1} + \cdots + a_1^*(t)p + 1, \quad (8.2)$$

$$B^*(p, t) = b_m^*(t)p^m + b_{m-1}^*(t)p^{m-1} + \cdots + b_1^*(t)p + b_0^*(t), \quad (8.3)$$

with $a_n^* \neq 0$ and $n \geq m$. These polynomials are assumed to be coprime, i.e., for any fixed t along the parameter trajectory, they do not share roots in p .

The system in (8.1) is represented in its unfactored function form. Any non-additive transfer function that represents (8.1) can also be described as a sum of time-variant systems of lower order. Such decomposition is called an additive form of (8.1), which is described by

$$x_i(t) = G_i^*(p, t)u(t), \quad (8.4a)$$

$$x(t) = \sum_{i=1}^K x_i(t), \quad (8.4b)$$

where $G_i^*(p, t) = B_i^*(p, t)/A_i^*(p, t)$, and $A_i^*(p, t)$ and $B_i^*(p, t)$ have the same form as (8.2) and (8.3), but have orders n_i and m_i with $a_{i,n_i}^* \neq 0$, $n_i \geq m_i$, and

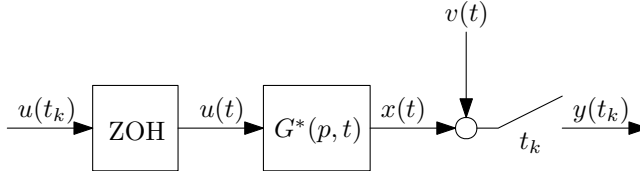


Figure 8.1: Block diagram for the open-loop setting studied in this chapter.

$\sum_{i=1}^K n_i = n$. That is,

$$A_i^*(p, t) = a_{i, n_i}^*(t)p^{n_i} + a_{i, n_i-1}^*(t)p^{n_i-1} + \cdots + a_{i, 1}^*(t)p + 1, \quad (8.5)$$

$$B_i^*(p, t) = b_{i, m_i}^*(t)p^{m_i} + b_{i, m_i-1}^*(t)p^{m_i-1} + \cdots + b_{i, 1}^*(t)p + b_{i, 0}^*(t). \quad (8.6)$$

Without loss of generality, it is assumed that the polynomials $A_i^*(p, t)$ are jointly coprime and anti-monic, i.e., their constant coefficient is fixed to one. In addition, for each separate submodel to be identifiable, the system is parameterized such that at most one subsystem $G_i^*(p, t)$ is biproper. An approach to determine an additive decomposition of (8.1) is by calculating the partial fraction decomposition of $B^*(p, t)/A^*(p, t)$.

The polynomials $A_i^*(p, t)$ and $B_i^*(p, t)$ are jointly described by the parameter vector

$$\theta_i^*(t) = [a_{i, 1}^*(t) \quad a_{i, 2}^*(t) \quad \cdots \quad a_{i, n_i}^*(t) \quad b_{i, 0}^*(t) \quad b_{i, 1}^*(t) \quad \cdots \quad b_{i, m_i}^*(t)]^\top, \quad (8.7)$$

and

$$\beta^*(t) := [\theta_1^{*\top}(t) \quad \theta_2^{*\top}(t) \quad \cdots \quad \theta_K^{*\top}(t)]^\top. \quad (8.8)$$

A noisy output measurement is retrieved at equidistantly-spaced time instants $t = t_k$, see Figure 8.1, i.e.,

$$y(t_k) = x(t_k) + v(t_k), \quad (8.9)$$

where v is a stochastic process of zero mean and finite variance σ^2 .

If $G^*(p, t)$ is in closed loop, see Figure 8.2, it is assumed that the system is driven by a reference signal $r(t_k)$ and the control input is determined through the control law

$$u(t_k) = C_d(q)(r(t_k) - y(t_k)), \quad (8.10)$$

where $C_d(q)$ denotes the (discrete-time) controller transfer function, and q denotes the forward shift operator. It is assumed that the input sequence $u(t_k)$ is interpolated via a zero-order hold prior to exciting the system. In closed loop, the input is described as

$$u(t_k) = \tilde{r}(t_k) - \tilde{v}(t_k), \quad (8.11)$$

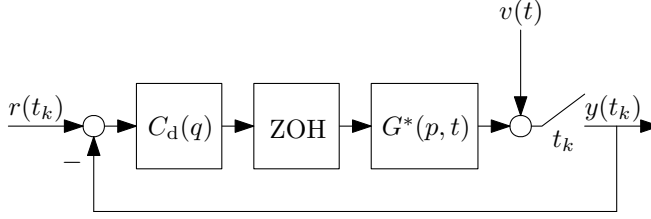


Figure 8.2: Block diagram for the closed-loop setting studied in this chapter.

where

$$\tilde{r}(t_k) := S_{uo}^*(q, t_k)r(t_k), \quad \tilde{v}(t_k) := S_{uo}^*(q, t_k)v(t_k), \quad (8.12)$$

with $S_{uo}^*(q, t_k)$ being the sensitivity function $S_{uo}^*(q, t_k) = C_d(q)/[1 + G_d^*(q, t_k)C_d(q)]^{-1}$, and $G_d^*(q, t_k)$ being the zero-order-hold equivalent discrete-time system of $G^*(p, t) = \sum_{i=1}^K G_i^*(p, t)$. Explicit expressions for computing $G_d^*(q, t_k)$ are obtained when changes in the parameters in $A^*(p, t)$ and $B^*(p, t)$ occur only at the sampling time instants t_k , see Tóth [250, Chap. 6.3.1] for more details.

8.2.2 Problem formulation

Consider the CT LTV system with input $u(t)$ and output $y(t)$ in (8.4). The data-generating system is described by (8.9) in the open-loop setting. In case the system is closed-loop controlled, the data-generating system additionally includes (8.10) and setpoint $r(t)$.

It is assumed that the plant is represented by the model set

$$x_i(t) = G_i(p, t, \theta_i)u(t), \quad (8.13a)$$

$$x(t) = \sum_{i=1}^K x_i(t), \quad (8.13b)$$

$$y(t_k) = x(t_k) + v(t_k). \quad (8.13c)$$

Here $G_i(p, t, \theta_i) = B_i(p, t, \theta_i)/A_i(p, t, \theta_i)$, where the polynomials $A_i(p, t, \theta_i)$ and $B_i(p, t, \theta_i)$ have the same structure as (8.5) and (8.6) with $\theta_i(t) \in \mathbb{R}^{n_{\theta_i}}$, and $n_{\theta_i} = n_i + m_i + 1$. The estimate of the total additive form is described by $\beta(t) \in \mathbb{R}^{n_\beta}$, which has the same structure as (8.8) and where $n_\beta = \sum_{i=1}^K (n_i + m_i) + K$.

The considered problem is to recursively estimate $\beta(t)$ which characterizes the model structure given by (8.13), based on the input and output data $\{u(t_k), y(t_k)\}_{k=1}^N$. Here, N is the number of samples available at time t_N for the open-loop scenario, while $\{r(t_k)\}_{k=1}^N$ is also known in the closed-loop setting.

Next, the primary benefits of additive parameterizations are presented, supported by multiple motivating examples.

8.3 Parsimony and interpretability in continuous-time system identification

The estimation in an additive form has several benefits including the following.

1. Direct continuous-time identification methods may suffer from a lack of parsimony when the sum of transfer functions of particular relative degrees is estimated [108].
2. The parameters in an additive structure may be more interpretable compared to an unfactored transfer function.
3. Typically, derivatives of the input and output data are required for recursive estimation. Considering an additive structure can significantly reduce the order of the derivatives to be estimated from sampled data.

First, the advantages regarding parsimony and interpretability are described. Later, in Section 8.4.2, Remark 8.6, the additional advantage regarding the derivative order is highlighted.

Direct continuous-time identification methods such as the SRIVC method may suffer from a lack of parsimony when identifying a system that consists of the sum of transfer functions. The following proposition quantifies the additional parameters being estimated when opting to estimate (8.1) with a model structure $G(p) = B(p)/A(p)$ of relative degree r instead of separately estimating the parameters for each transfer function $B_i(p)/A_i(p)$.

Proposition 1. [108] *Consider the system in (8.1), and the model structure $G(p) = \sum_{i=1}^K B_i(p)/A_i(p)$. Opting for the model structure $G(p) = B(p)/A(p)$ during identification (with the minimal relative degree r encompassing the true system) results in a lack of parsimony for the latter model structure if and only if*

$$\sum_{i=1}^K r_i - r > K - 1, \quad (8.14)$$

where $r_i = n_i - m_i$ is the relative degree of $B_i(p)/A_i(p)$. The surplus in (8.14) is the number of additional parameters considered by the model structure $G(p) = B(p)/A(p)$.

An immediate implication derived from Proposition 1 is that when $r = 0$ or $r = 1$, the model structure $G(p) = B(p)/A(p)$ suffers from a lack of parsimony if a transfer function $G_i^*(p)$ exists in its partial fraction expansion (8.1) with a relative degree greater than one. Two examples of this property are given below.

Example 8.1. *Consider the system*

$$G^*(p) = \frac{b_{1,0}^*}{a_{1,1}^*p + 1} + \frac{b_{2,0}^*}{a_{2,2}^*p^2 + a_{2,1}^*p + 1},$$

where the $a_{i,j}^*$ values are strictly positive, and the $b_{i,0}^*$ are non-zero. Only 5 parameters must be estimated if the following model structure is used:

$$G(p) = \frac{b_{1,0}}{a_{1,1}p + 1} + \frac{b_{2,0}}{a_{2,2}p^2 + a_{2,1}p + 1}. \quad (8.15)$$

Alternatively, if the consideration of this partial fraction decomposition is disregarded and the decision is made to employ the model structure

$$G(p) = \frac{b_2p^2 + b_1p + b_0}{a_3p^3 + a_2p^2 + a_1p + 1}$$

with no constraints on the parameter values, then 6 parameters must be estimated. This model structure leads to a lack of parsimony compared to the one in (8.15). $\triangle$

Example 8.2. Consider the system

$$G^*(p) = \frac{b_0^*}{a_2^*p^2 + a_1^*p + 1},$$

where a_1^* and a_2^* are strictly positive, and $a_1^* > 2\sqrt{a_2^*}$. Hence, $G^*(p)$ has two real-valued poles. If the following partial fraction decomposition of $B^*(p)/A^*(p)$ is considered

$$G(p) = \frac{b_{1,0}}{a_{1,1}p + 1} + \frac{b_{2,0}}{a_{2,1}p + 1},$$

then 4 parameters must be estimated instead of 3. This additive representation does not directly take into account the relationship between the numerator terms $b_{1,0}$ and $b_{2,0}$, leading to a lack of parsimony. $\triangle$

The following example shows a mechanical system in one of its canonical forms. In addition to parsimony for these type of systems, also the interpretability of the parameters is examined.

Example 8.3. Mechanical systems are typically described as a sum of modes, i.e.,

$$G^*(p) = \sum_{i=1}^K \frac{b_{i,0}^*}{p^2/\omega_i^2 + 2(\zeta_i/\omega_i)p + 1}, \quad (8.16)$$

where $\omega_i > 0$ and $\zeta_i > 0$ denote the natural frequency and the damping coefficient of the i th subsystem [91]. These systems are described by $3K$ parameters if the following model structure is used:

$$G(p) = \sum_{i=1}^K \frac{b_{i,0}}{a_{i,2}p^2 + a_{i,1}p + 1}. \quad (8.17)$$

On the contrary, if the partial fraction decomposition is overlooked and the decision is made to adopt the widely utilized model structure of order n and relative degree two

$$G(p) = \frac{b_{n-2}p^{n-2} + b_{n-3}p^{n-3} + \dots + b_0}{a_np^n + a_{n-1}p^{n-1} \dots + a_1p + 1}, \quad (8.18)$$

with $n = 2K$ and no constraints on the parameter values, then a total of $4K - 1$ parameters must be estimated. This model structure leads to a lack of parsimony compared to the one in (8.17) if $K > 1$, with a further degradation of parsimony for larger values of K . Additionally, the parameters in (8.17) are physically interpretable as these directly relate to the natural frequencies ω_i and the damping coefficients ζ_i . This interpretability is hidden and mostly lost in (8.18). $\triangle$

Interpretability of the parameters, as illustrated in the latter example, is highly favourable in many engineering applications. For instance, in the field of fault diagnosis, this interpretability greatly simplifies the task of fault isolation, as faults often manifest in a specific submodel. Estimating the system in an unfactored form leads to variations in most parameters, while estimating it in an additive form results in variations solely in the directly related parameters.

8.4 Recursive estimation of continuous-time systems

This section provides a recursive solution to estimate additive models. A key optimization tool that is considered in the approach is the block-descent method, which is examined in the next subsection. Then, the recursive estimators are presented for both open and closed-loop settings. Additionally, a summary of the algorithm is presented, adaptations for marginally stable systems are given, and practical aspects are described.

8.4.1 Block coordinate descent

Towards the goal of computing a recursive estimator for β^* , consider the minimization problem

$$\min_{\substack{\theta_i \in \Omega_i \\ i=1, \dots, K}} V_N(\beta), \quad (8.19)$$

where $\Omega_i \subset \mathbb{R}^{n_{\theta_i}}$ is a compact set where the true parameters of the i th subsystem are assumed to be. The cost V_N , using the N available samples at t_N , is the weighted least squares output error cost function

$$V_N(\beta) = \sum_{k=1}^N \alpha_{N-k} \left[y(t_k) - \sum_{i=1}^K G_i(p, t, \theta_i) u(t_k) \right]^2, \quad (8.20)$$

where the notation $G_i(p, t, \theta_i)u(t_k)$ means that the input $u(t)$ is filtered through the continuous-time transfer function $G_i(p, t, \theta_i)$ and later evaluated at $t = t_k$. The function α is a non-negative weighting function of the form

$$\alpha_k = \lambda^k, \quad \lambda \in (0, 1]. \quad (8.21)$$

Note that α satisfies the multiplicative property $\alpha_{k+1} = \lambda \alpha_k$, with $\alpha_0 = 1$. The parameter λ is often referred to as the forgetting factor [161].

The approach for solving (8.19) consists of iteratively minimizing the cost with respect to θ_i while leaving the other variables fixed. Algorithm 1 describes the general procedure that is solved, typically after every incoming sample, known as *block coordinate descent* [163].

Algorithm 1 Block coordinate descent

- 1: Choose an initial parameter vector θ_i^1 for each i
 - 2: **for** $l = 1, 2, \dots$ **do**
 - 3: **for** $i = 1, \dots, K$ **do**
 - 4: $\theta_i^{l+1} \leftarrow \arg \min_{\theta_i \in \Omega_i} V_N(\theta_1^{l+1}, \dots, \theta_{i-1}^{l+1}, \theta_i, \theta_{i+1}^l, \dots, \theta_K^l)$
 - 5: **end for**
 - 6: **end for**
-

Let $\beta^l := [(\theta_1^l)^\top, \dots, (\theta_K^l)^\top]^\top \in \Omega \subset \mathbb{R}^{n_\beta}$, with $\Omega = \prod_{i=1}^K \Omega_i$ being the parameter space. Each iteration of the algorithm (in l) can therefore be written as $\beta^{l+1} = \mathcal{A}(\beta^l)$, where $\mathcal{A}: \mathbb{R}^{n_\beta} \rightarrow \mathbb{R}^{n_\beta}$ is the composite mapping $\mathcal{A} = \mathcal{T} \circ \mathcal{C}^K \circ \mathcal{T} \circ \mathcal{C}^{K-1} \circ \dots \circ \mathcal{T} \circ \mathcal{C}^1$. This notation is adopted from [163]. Here, the choice function is defined as $\mathcal{C}^i(\beta) := (\beta, i)$, and the optimization function $\mathcal{T}(\beta, i) := (\theta_1, \dots, \theta_{i-1}, \bar{\theta}_i, \theta_{i+1}, \dots, \theta_K)$, with $\bar{\theta}_i$ being defined as $\bar{\theta}_i = \arg \min_{\theta_i \in \Omega_i} V_N(\theta_1, \dots, \theta_i, \dots, \theta_K)$. The following result concerns the convergence of Algorithm 1 to a stationary point of the cost (8.19).

Theorem 8.4 (Global convergence of Algorithm 1). *If the search along any coordinate direction θ_i yields a unique minimum point of V_N , then the limit of any convergent subsequence of $\{\beta^l\}$ obtained from $\beta^{l+1} = \mathcal{A}(\beta^l)$ belongs to the set of fixed points $\Gamma_N = \{\beta \in \Omega: \nabla V_N(\beta) = 0\}$.*

Proof. See, e.g., Luenberger and Ye [163, Section 8.9]. □

Remark 8.5. *The result in Theorem 8.4 still holds if the unique minimum assumption is replaced with the requirement that V_N decreases at each coordinate search. In this case it is sufficient to find a parameter vector $\bar{\theta}_i$ that reduces the cost V_N instead of minimizing it.*

The block coordinate descent algorithm described in Algorithm 1 requires a way to compute $\mathcal{T}(\beta, i)$ at each iteration, for each $i = 1, 2, \dots, K$. That is, given

the data up until t_N , compute

$$\theta_i^{l+1} = \arg \min_{\theta_i \in \Omega_i} \sum_{k=1}^N \alpha_{N-k} \left[y(t_k) - \sum_{j=1}^{i-1} G_j(p, t, \theta_j^{l+1}) u(t_k) - \sum_{j=i+1}^K G_j(p, t, \theta_j^l) u(t_k) - G_i(p, t, \theta_i) u(t_k) \right]^2 \quad (8.22)$$

for $i = 1, 2, \dots, K$. For fixed values of $\{\theta_j^{l+1}\}_{j=1}^{i-1}$ and $\{\theta_j^l\}_{j=i+1}^K$. For a sufficiently large compact parameter space Ω_i , the optimization problem in (8.22) becomes a nonlinear least-squares problem that lends itself to iterative solution methods. To this end, define the residual output of each submodel as

$$\tilde{y}_i(t_k) := y(t_k) - \sum_{j=1}^{i-1} G_j(p, t, \theta_j^{l+1}) u(t_k) - \sum_{j=i+1}^K G_j(p, t, \theta_j^l) u(t_k).$$

Substitution of this residual output in (8.22) gives

$$\theta_i^{l+1} = \arg \min_{\theta_i \in \Omega_i} \sum_{k=1}^N \alpha_{N-k} \left[\tilde{y}_i(t_k) - G_i(p, t, \theta_i) u(t_k) \right]^2. \quad (8.23)$$

Next is shown how this is solved through recursive SRIVC.

8.4.2 Recursive SRIVC for additive open-loop systems

In the linear and time-invariant case, when solved iteratively, the optimization problem in (8.23) leads to a local minimum for the complete optimization problem (8.19) [108]. Each optimization problem in (8.23) takes the form of minimizing the weighted sum of squares of the residual $\varepsilon_i(t_k, \theta_i) = \tilde{y}_i(t_k) - G_i(p, t, \theta_i) u(t_k)$. This pseudolinear regression takes the general form

$$\tilde{y}_{i,f}(t_k, \theta_i) = \varphi_{i,f}^\top(t_k, \theta_i) \theta_i(t_k) + \varepsilon_i(t_k, \theta_i). \quad (8.24)$$

This generic form admits the refined instrumental variable iterations [281]. Essentially, the refined instrumental variable iterations provide a sequence which, under mild conditions, converges to a point $\bar{\theta}_i$ that satisfies the first order optimality condition of the cost in (8.23), i.e.,

$$\sum_{k=1}^N \alpha_{N-k} \frac{\partial \varepsilon_i(t_k, \theta_i)}{\partial \theta_i} \bigg|_{\theta_i = \bar{\theta}_i} \varepsilon_i(t_k, \bar{\theta}_i) = 0.$$

The instrumental variable refinements, from now denoted as SRIVC iterations, are denoted using the iterate counter s , as

$$\begin{aligned} \theta_{i,s+1}^{l+1} = & \left[\sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_k, \theta_{i,s}^{l+1}) \right]^{-1} \\ & \times \left[\sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \tilde{y}_{i,f}(t_k, \theta_{i,s}^{l+1}) \right]. \end{aligned} \quad (8.25)$$

The regressors are denoted by ϕ and the instruments are denoted by $\hat{\phi}$. Appendix 8.A shows how (8.23) fits the pseudolinear regression framework with (8.24) and how this leads to the RSRIVC iterations in (8.25). In (8.25), the subscripts i and l refer to the current block-coordinate descent iterate. The filtered residual output, regressor vector and instrument vector, indicated with subscript f, are respectively given by

$$\tilde{y}_{i,f}(t_k, \theta_{i,s}^{l+1}) = \frac{1}{A_i(p, \theta_{i,s}^{l+1})} \tilde{y}_i(t_k), \quad (8.26)$$

$$\begin{aligned} \varphi_{i,f}(t_k, \theta_{i,s}^{l+1}) = & \left[\frac{-p}{A_i(p, \theta_{i,s}^{l+1})} \tilde{y}_i(t_k) \quad \dots \quad \frac{-p^{n_i}}{A_i(p, \theta_{i,s}^{l+1})} \tilde{y}_i(t_k) \right. \\ & \left. \frac{1}{A_i(p, \theta_{i,s}^{l+1})} u(t_k) \quad \dots \quad \frac{p^{m_i}}{A_i(p, \theta_{i,s}^{l+1})} u(t_k) \right]^\top, \end{aligned} \quad (8.27)$$

$$\begin{aligned} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) = & \left[\frac{-p}{A_i(p, \theta_{i,s}^{l+1})} \hat{x}_i(t_k) \quad \dots \quad \frac{-p^{n_i}}{A_i(p, \theta_{i,s}^{l+1})} \hat{x}_i(t_k) \right. \\ & \left. \frac{1}{A_i(p, \theta_{i,s}^{l+1})} u(t_k) \quad \dots \quad \frac{p^{m_i}}{A_i(p, \theta_{i,s}^{l+1})} u(t_k) \right]^\top, \end{aligned} \quad (8.28)$$

where $\hat{x}_i(t_k)$ is the estimate of $x_i(t_k)$. This estimate is based on the latest parameter estimate of the previous time-sample, denoted by $\bar{\theta}_i(t_{k-1})$, i.e.,

$$\hat{x}_i(t_k) = \frac{B_i(p, \bar{\theta}_i(t_{k-1}))}{A_i(p, \bar{\theta}_i(t_{k-1}))} u(t_k).$$

To clarify, the refinement step refers to updating the denominator polynomial of the prefiltering step in (8.26) to (8.28). Hence, the filtered output, regressor vector, and instrument vector, are updated based on the estimates obtained from the previous iterate s . The instrument vector is similar to the regressor vector but uses $\hat{x}_i$ instead of $\tilde{y}_i$, i.e., an estimate of the output of the submodel instead of using the noisy measured variable.

Remark 8.6. The regressor (8.27) and instrument (8.28) require the computation of derivatives up to order m_i for the input and n_i for the residual output.

Note that the order of the required derivatives is significantly lower compared to the case where a single unfactored transfer function would have been used. In this scenario, given that $\sum_{i=1}^K n_i = n$, obtaining the n th order derivative of the output is necessary for standard recursive continuous-time identification methods, along with a substantially higher derivative of the input.

Remark 8.7. Note that the expressions in (8.25) to (8.28) give the standard SRIVC iterations when $\tilde{y}(t_k) = y(t_k)$ (i.e., when $K = 1$ and all the modes of the composite transfer function are estimated jointly) and $\alpha_{N-k} = 1$ for all integer k less than N (i.e., when there is no forgetting factor).

Remark 8.8. The expressions provided in (8.26), (8.27) and (8.28) require prefiltering the data, relying implicitly on the commutativity property given by

$$cG(p)x(t) = G(p)[cx(t)], \quad (8.29)$$

where c is a constant. It is not recommended to prefilter the data by transfer functions when the parameters exhibit rapid temporal variations, since the property (8.29) does not hold in general when c is time-varying [250]. However, it is important to note that (8.29) is a useful approximation in cases where the parameters exhibit slow temporal variations.

Instead of computing (8.25), an equivalent recursion is derived which does not involve the summation operator over all available samples and allows to achieve an update of the estimate after every incoming time sample. This recursion involves computing

$$K_{i,s+1}^{l+1}(t_N) = \bar{P}_i(t_{N-1})\hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \times (\lambda + \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1})\bar{P}_i(t_{N-1})\hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}))^{-1}, \quad (8.30)$$

$$\theta_{i,s+1}^{l+1}(t_N) = \bar{\theta}_i(t_{N-1}) + K_{i,s+1}^{l+1}(t_N) \times (\tilde{y}_{i,f}(t_N, \theta_{i,s}^{l+1}) - \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1})\bar{\theta}_i(t_{N-1})), \quad (8.31)$$

and

$$P_{i,s+1}^{l+1}(t_N) = \frac{1}{\lambda} (\bar{P}_i(t_{N-1}) - K_{i,s+1}^{l+1}(t_N)\varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1})\bar{P}_i(t_{N-1})), \quad (8.32)$$

where $\bar{\theta}_i(t_{N-1})$, and $\bar{P}_i(t_{N-1})$ denote the last parameter estimate and inverse of the matrix being inverted in (8.25) at the previous time instance t_{N-1} after the $\mathcal{A}$ -iterations over l and i , and after the SRIVC iterations using s . The derivation of this recursion is presented in Appendix 8.B. Note that $\bar{P}_i(t_{N-1})$ is interpreted as an approximate covariance matrix of the resulting estimate, see [278] and [192] for more details.

Next it is analyzed under which conditions the additive RSRIVC method yields a consistent estimator of the continuous-time system. To this end, consider the following assumptions to facilitate the analysis, theorem, and related remarks.

Assumption 8.9 (BIBO stability). *The true system and the estimated models are uniformly bounded-input, bounded-output (BIBO) stable, i.e., for any initial time instant t_0 and any bounded input signal, the output sequence $\{y(t_k)\}_{k \geq 0}$ is bounded [214].*

Assumption 8.10 (Persistence of excitation). *The input sequence is persistently exciting of order no less than $2n$.*

Assumption 8.11 (Coprimeness and sampling frequency). *The additive subsystems pertaining to the true system and the models obtained through the descent iterations do not have zero-pole cancellations. Furthermore, the sampling frequency is more than twice the largest positive imaginary part of the zeros of $\prod_{i=1}^K A_i(p, \theta_{i,s}^l) A_i^*(p)$. The subscript s and superscript l relate to the SRIVC iterations and block-coordinate descent iterations, respectively, defined in Section 8.4.*

Assumption 8.12 (Independence and stationarity). *The input sequence $\{u(t_k)\}$ is quasi-stationary [161, p. 34], the disturbance $\{v(t_l)\}$ is stationary, and they are mutually independent for all integers k and l .*

Assumption 8.13 (Parameter trajectories). *The changes in the parameters occur only at the sampling instants t_k , i.e., the parameter trajectories are piecewise constant.*

Theorem 8.14. *Consider the proposed estimator for the open-loop setting in Figure 8.1 with $\alpha_k = 1$ for all k , and suppose Assumptions 8.9 to 8.13 hold. In addition, assume that the SRIVC iterations of the estimator for all N sufficiently large converge, where the converged parameter is denoted by $\theta_{i,\infty}^{l+1}$. Then, if both the proposed model and system converge as $N \rightarrow \infty$ to linear and time-invariant transfer functions, then their limits are equal with probability 1 and generically consistent² with respect to the limiting system and model parameters. In the remainder of this work, the following is assumed primarily to facilitate the analysis.*

The proof is provided in Appendix 8.C.

Remark 8.15. *Note that consistency is only considered for $\alpha_k = 1$ for all k . This is a special case of (8.21). In case (8.21) is used with a different constant-rate-forgetting factor $\lambda < 1$, the effect of noise does not disappear for $N \rightarrow \infty$. Therefore, considering more data does not reduce the influence of measurement noise making the estimator not consistent. Variable-rate-forgetting factors may be considered, ensuring consistent estimators (see [29] for details); however, these considerations are beyond the scope of this work.*

²A statement s , which depends on the elements ρ of some open set $\Omega \subseteq \mathbb{R}^n$, is *generically true with respect to Ω* [234] if the set $\{\rho \in \Omega | s(\rho) \text{ is not true}\}$ has Lebesgue measure zero in Ω .

8.4.3 Summary iterative recursive estimation algorithm

The proposed additive RSRIVC algorithm involves nested iterations in l , i , and s . To facilitate implementation, the nested iterations are discussed in this subsection.

First of all, the recursive nature allows to update the estimate after every incoming sample. Therefore, these iterations will be performed at the discrete times t_k , where k denotes the count in the discrete-time domain. At each of these points in time, the $\mathcal{A}$ -iterations are performed M_l times, see Section 8.4.1, i.e., the block-coordinate method runs through each of K additive models for M_l times. The current additive model is denoted by i and the $\mathcal{A}$ -iteration is denoted by l . Finally, within each of these parameter updates, M_s SRIVC iterations are performed, with the current SRIVC iterate being indicated by s .

To summarize, compute the recursive formulas in (8.30), (8.31) and (8.32) at each time step and for every additive model G_i to solve the additive model identification problem in (8.23). This way, the proposed method solves (8.23) recursively. The algorithm is presented in Algorithm 2. Note that the regressor vector has an explicit dependence on the parameter vector of the other models being estimated; this dependence is implicit in the equations via the expression for the residual $\tilde{y}_i$. A block-diagram of the block-coordinate descent RSRIVC algorithm for additive systems is shown in Figure 8.3. Two distinct blocks are highlighted in this diagram, which refer to the SRIVC and the block-coordinate descent step. These are executed after every incoming sample. Optionally, a stability test is included which is further examined in Section 8.4.6.

Remark 8.16. *Note that the SRIVC refinements in s are optional. In case the refinements are omitted, the optimization problem (8.23) is not being solved until convergence at every time-step, but rather a quasi-Newton descent algorithm of the form (8.25) is being implemented [278]. Despite this, the iterations stem from the fact that solving (8.23) and iterating over the additive models is a viable way to derive a model that fully minimizes the cost in (8.19).*

Remark 8.17. *While not inherently restrictive to the method, it is practically advantageous if parameters change gradually. In general, slower parameter variation improves the effectiveness of the methods by reducing computational burden, improving parameter adaptation efficiency, and robustness.*

Algorithm 2 Recursive estimator for open-loop continuous-time systems in additive form

- 1: **Initialization:** Set M_l , M_s , λ . Initialize $\bar{\theta}_i(t_{N_1})$, and $\bar{P}_i(t_{N_1})$ for each $i = 1, \dots, K$, using the first part of the input-output data $\{u(t_k), y(t_k)\}_{k=-N_{\text{init}}}^0$ with any linear and time-invariant continuous-time system identification method. With $\bar{\theta}_i(t_0)$ and the input-output data, initialize $\varphi_{i,f}$, $\hat{\varphi}_{i,f}$, and $\tilde{y}_{i,f}$.
 - 2: **for** $k = 1, \dots$ **do**
 - 3: $\theta_i^1(t_k) \leftarrow \bar{\theta}_i(t_{k-1})$ for all i
 - 4: **for** $l = 1, \dots, M_l$ **do**
 - 5: **for** $i = 1, \dots, K$ **do**
 - 6: $\theta_{i,0}^{l+1}(t_k) \leftarrow \theta_i^l(t_k)$
 - 7: **for** $s = 0, 1, \dots, M_s$ **do**
 - 8: Parameter $\theta_{i,s+1}^{l+1}(t_k)$ update:

$$K_{i,s+1}^{l+1}(t_k) \leftarrow \frac{\bar{P}_i(t_{k-1})\hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1})}{\lambda + \varphi_{i,f}^\top(t_k, \theta_{i,s}^{l+1})\bar{P}_i(t_{k-1})\hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1})},$$

$$\theta_{i,s+1}^{l+1}(t_k) \leftarrow \bar{\theta}_i(t_{k-1}) + K_{i,s+1}^{l+1}(t_k) (\tilde{y}_{i,f}(t_k, \theta_{i,s}^{l+1}) - \varphi_{i,f}(t_k, \theta_{i,s}^{l+1})\bar{\theta}_i(t_{k-1})),$$

$$P_{i,s+1}^{l+1}(t_k) \leftarrow \frac{1}{\lambda}\bar{P}_i(t_{k-1}) - \frac{1}{\lambda}K_{i,s+1}^{l+1}(t_k)\varphi_{i,f}^\top(t_k, \theta_{i,s}^{l+1})\bar{P}_i(t_{k-1}).$$
 - 9: **end for**
 - 10: $\theta_i^{l+1}(t_k) \leftarrow \theta_{i,M_s+1}^{l+1}(t_k)$
 - 11: $P_i^{l+1}(t_k) \leftarrow P_{i,M_s+1}^{l+1}(t_k)$
 - 12: **end for**
 - 13: **end for**
 - 14: $\bar{\theta}_i(t_k) = \theta_i^{M_l+1}(t_k)$ for all i
 - 15: $\bar{P}_i(t_k) = P_i^{M_l+1}(t_k)$ for all i
 - 16: **end for**
 - 17: **Output:** the parameter estimates $\{\bar{\theta}_i(t_k)\}_{k \geq 1}$.
-

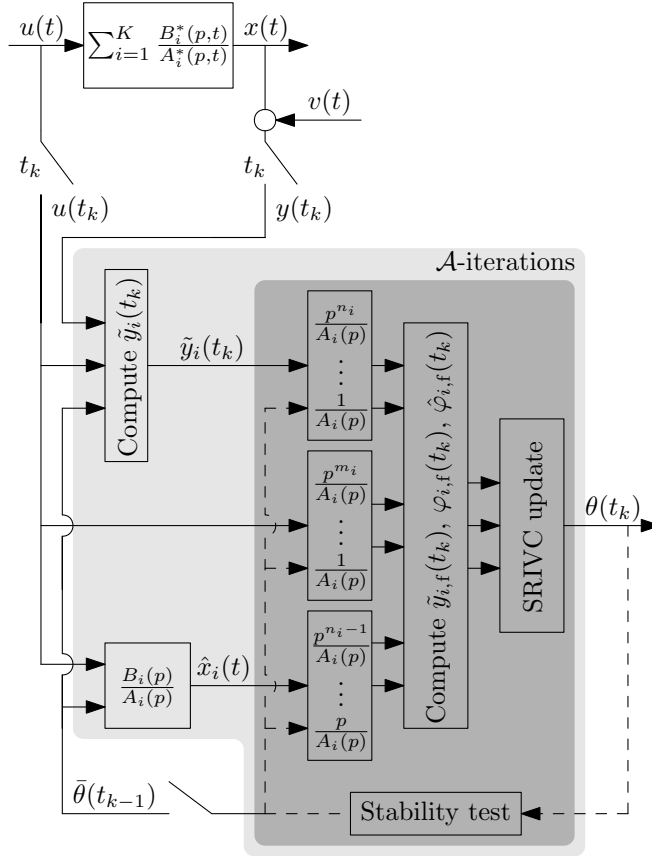


Figure 8.3: Block diagram of the block-coordinate descent RSRIVC for open-loop systems, summarized by Algorithm 2. The section featuring SRIVC iterations is highlighted (■), see (8.25) to (8.28), alongside the block containing the $\mathcal{A}$ -iterations (■), described in Section 8.4.1.

8.4.4 Closed-loop case

The identical recursion as in (8.30), (8.31) and (8.32) applies to closed-loop scenario, albeit with a different instrument vector. Employing an output error cost for system identification is known to introduce asymptotic bias in the presence of output measurement noise [256]. Consequently, instrumental variable methods are used in the closed-loop setting to mitigate this bias [98, 106]. Instead of (8.23), the instrumental variable solution is employed which is given by

$$\theta_i^{l+1} \in \underset{\theta_i \in \Omega_i}{\text{sol}} \left\{ \sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_i) (\tilde{y}_{i,f}(t_k, \theta_i) - \varphi_{i,f}^\top(t_k, \theta_i) \theta_i(t_k)) = 0 \right\},$$

with $\hat{\varphi}_{i,f}$ being the filtered instrument vector related to additive model i . Users have the flexibility to choose from various instrument vectors [234]. In this context, the instrument is constructed as a noiseless version of the regressor, a choice recognized in similar identification contexts for yielding minimal asymptotic covariance among all instrumental variable estimators [24, 97, 112, 174]. To this end, the instrument (8.28) is rewritten as

$$\hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) = \left[\frac{-p}{A_i(p, \theta_{i,s}^{l+1})} \frac{B_i(p, \bar{\theta}_i(t_{k-1}))}{A_i(p, \bar{\theta}_i(t_{k-1}))} \cdots \frac{-p^{n_i}}{A_i(p, \theta_{i,s}^{l+1})} \frac{B_i(p, \bar{\theta}_i(t_{k-1}))}{A_i(p, \bar{\theta}_i(t_{k-1}))} \right. \\ \left. \frac{1}{A_i(p, \theta_{i,s}^{l+1})} \cdots \frac{p^{m_i}}{A_i(p, \theta_{i,s}^{l+1})} \right]^\top z(t_k), \quad (8.33)$$

however, instead of $z(t_k) = u(t_k)$, the noiseless input is estimated as $z(t_k) = S_{uo}(q, t_k)r(t_k)$, where the sensitivity function $S_{uo}(q, t_k) = C_d(q)/[1 + C_d(q)G_d(q, t_k)]^{-1}$. Hence, the instrument now depends on an estimate of the control input, computed by means of a model of the sensitivity and the known setpoint $r(t_k)$.

8.4.5 Marginally stable systems

Mechatronic position systems, frequently encountered in engineering applications, lend themselves to a decomposition comprising both rigid body and flexible modes. Notably, the rigid body modes are considered as double integrators, leading to eigenvalues precisely situated at the origin. The denominator polynomial of such marginally stable system cannot be represented in an anti-monic form as in (8.5) and (8.6).

The proposed approach is extended to the estimation of marginally stable systems as follows. Consider the system (8.4), but where the first submodel has ℓ poles at the origin. In other words, let $G_1^*(p, t) = B_1^*(p, t)/[p^\ell A_1^*(p, t)]$, where $A_1^*(p, t)$ and $B_1^*(p, t)$ are co-prime, and have the same form as in (8.2) and (8.3). For either open or closed-loop variants of the proposed approach, the computation of the gradient of each submodel with respect to their respective parameter vector is required. For $G_1(p)$, this computation leads to the following instrument vector.

$$\hat{\varphi}_{1,f}(t_k, \theta_{1,s}^{l+1}) = \left[\frac{-p}{p^\ell A_1(p, \theta_{1,s}^{l+1})} \frac{B_1(p, \bar{\theta}_1(t_{k-1}))}{A_1(p, \bar{\theta}_1(t_{k-1}))} \cdots \frac{-p^{n_1}}{p^\ell A_1(p, \theta_{1,s}^{l+1})} \frac{B_1(p, \bar{\theta}_1(t_{k-1}))}{A_1(p, \bar{\theta}_1(t_{k-1}))} \right. \\ \left. \frac{1}{p^\ell A_1(p, \theta_{1,s}^{l+1})} \cdots \frac{p^{m_1}}{p^\ell A_1(p, \theta_{1,s}^{l+1})} \right]^\top z(t_k), \quad (8.34)$$

where $z(t_k) = u(t_k)$ for the open-loop algorithm, and $z(t_k) = S_{uo}(q, t_k)r(t_k)$ for the closed-loop variant. On the other hand, the model residual retains the same

form as in (8.23), with the filtered residual output given by (8.24), but with the regressor vector now expressed as

$$\varphi_{1,f}(t_k, \theta_{1,s}^{l+1}) = \begin{bmatrix} \frac{-p}{A_1(p, \theta_{1,s}^{l+1})} \tilde{y}_1(t_k) & \cdots & \frac{-p^{n_1}}{A_1(p, \theta_{1,s}^{l+1})} \tilde{y}_1(t_k), \\ \frac{1}{p^\ell A_1(p, \theta_{1,s}^{l+1})} u(t_k) & \cdots & \frac{p^{m_1}}{p^\ell A_1(p, \theta_{1,s}^{l+1})} u(t_k) \end{bmatrix}^\top. \quad (8.35)$$

By computing the iterations in (8.25) with the first instrument and regressor vectors given by (8.34) and (8.35) respectively, a direct extension of the proposed method for identifying marginally stable systems in additive form is obtained.

Remark 8.18. *Assumption 8.9 is exclusively employed in the generic consistency proof for uniformly stable systems, excluding marginally stable systems, yet it does not constrain the proposed approach.*

8.4.6 Practical aspects for implementation

Next, several practical aspects of the additive RSRIVC implementation are investigated. First, the robustness of the algorithm is addressed. Subsequently, to alleviate the computational load associated with the approach, output subsampling is examined.

In the offline SRIVC method, ensuring model stability is vital, especially in challenging conditions like low Signal-to-Noise Ratio (SNR) or mismatched model orders [116]. Typically, stability is monitored and enforced throughout iterations [90]. In RSRIVC, where both prefilter and auxiliary model depend on preceding estimates, stability verification is more intricate. For slowly-varying systems, checking characteristic equation poles and reflecting right-half plane poles to the left suffices for confirming bounded-input, bounded-output stability under specific conditions [189]. Notably, in time-varying parameter models, uniform exponential stability is assured by assuming slow parameter changes and keeping system matrix eigenvalues on the left half plane. RSRIVC incorporates a projection algorithm to guarantee the stability of the prefilter and auxiliary model, reflecting unstable poles into the left half plane while preserving the magnitude characteristics. This approach addresses potential stability issues that may arise, particularly at the beginning of the data record.

The algorithm presented is theoretically sound, however, the recursion in (8.32) may not be numerically reliable [159, Chapter 11]. Since $P(t_k)$ is essentially computed by successive subtractions, round-off errors can accumulate and make $P(t_k)$ ill-conditioned. Other continuous-time recursive methods based on instrumental variables [190, 193] incorporate a covariance matrix $Q(t_k)$ that is added to $P(t_k)$ at each iteration. The proposed algorithm does not include this matrix since it is not compatible with the coordinate descent interpretation of

the iterations in Algorithm 2. However, from a practical perspective, including this matrix can help with the convergence of the parameters.

Subsampling data may be valuable to managing computational complexity without compromising significant accuracy. This facilitates the implementation of recursive algorithms in resource-constrained environments. Subsampling should be done cautiously to ensure that the subsampled data still captures the smallest timescale for a reliable estimate. Therefore, it is primarily applicable when the timescale of the system dynamics is considerably shorter than the sampling time. When subsampling the output, it is advised to take all input samples into account for the estimation of the filtered input and filtered auxiliary variables. Hence, the block-coordinate descent algorithm and additive RSRIVC iterations may be performed at this lower sampling rate. Furthermore, as hinted in Remark 8.16, it is not always necessary to perform multiple iterations. In case the parameters of the system vary slowly with respect to the sampling period, typically no refinements are necessary and a single $\mathcal{A}$ -iteration can achieve accurate estimates.

8.5 Numerical Simulation

In this section, the proposed method is tested in a practically relevant simulation example. Consider the following additive system

$$G^*(p, t) = \sum_{i=1}^3 \frac{b_{i,0}^*(t)}{a_{i,2}^*(t)p^2 + a_{i,1}^*(t)p + 1},$$

with initial DC-gains $[b_{1,0}^*(0) \ b_{2,0}^*(0) \ b_{3,0}^*(0)] = [3 \ 0.5 \ 1]$, and associated denominator parameters $[a_{1,2}^*(0) \ a_{2,2}^*(0) \ a_{3,2}^*(0)] = [0.25 \ 0.04 \ 0.0025]$, and $[a_{1,1}^*(0) \ a_{2,1}^*(0) \ a_{3,1}^*(0)] = [0.25 \ 0.01 \ 0.01]$. The output $y(t_k)$ is measured at a sampling frequency of 20 Hz and is contaminated by a zero-mean Gaussian white noise with a variance of 0.01. The system is operated in closed-loop with feedback controller

$$K(q) = \frac{0.02329q^2 - 0.2058q + 0.00454}{q^2 - q},$$

and the setpoint equals $r(t) = 3 \sin(0.005t) + \sin(2t) + \sin(5t) + \sin(17.5t)$. The algorithm is initialized with covariance matrices $\bar{P}_i(t_{N_1}) = \text{diag}(10^{-3}, 10^{-3}, 10^{-5})$, and forgetting factor $\lambda = 0.999$. The estimator is initialized at a random system with parameters $\bar{\theta}_i(t_{N_1}) = \theta_i^*(0)(1 + \mathcal{U}(-0.02, 0.02))$, where $\mathcal{U}(a, b)$ denotes a uniform distribution with lower limit a and upper limit b .

First, in the interval $t \in [100, 300]$ s, the parameters $a_{1,1}$, $a_{1,2}$, and $a_{2,1}$, are varied in such a way that the corresponding natural frequency ω_1 and damping

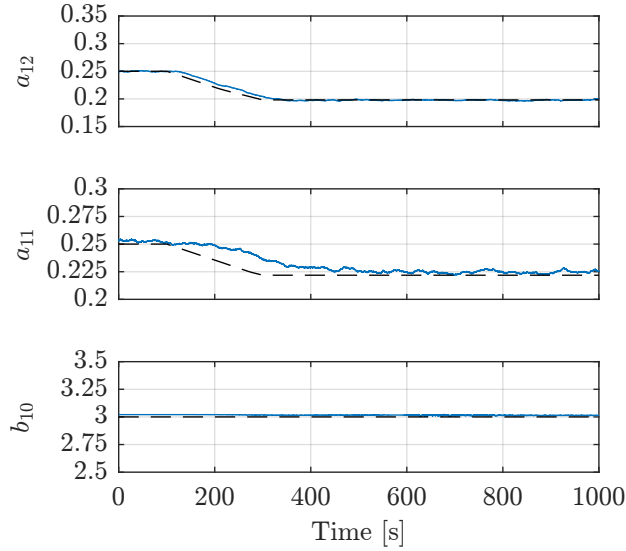


Figure 8.4: Parameter estimates (—) and true parameters (---) related to the first submodel with decreasing denominator parameters.

coefficient ζ_2 decrease, c.f., (8.16). Subsequently, in the interval $t \in [500, 800]$ s, the denominator parameters of the third model, i.e., $a_{3,2}$ and $a_{3,1}$ are varied.

Algorithm 2 is deployed to estimate the time-varying parameters, where no SRIVC iterations are performed and only a single coordinate descent iteration is performed at every incoming sample, i.e., the two loops in the algorithm are omitted as $s = 0$ and $M_t = 1$. Since the system is operating in closed loop, the instrument is chosen as (8.33). The obtained estimates and the true parameters are depicted for each of the submodels in Figures 8.4 to 8.6. The estimated model at $t = 50$ s and $t = 950$ s are depicted in Figure 8.7. From the figures it is concluded that no further refinements and gradient descent steps are required to achieve a accurate tracking of the parameters.

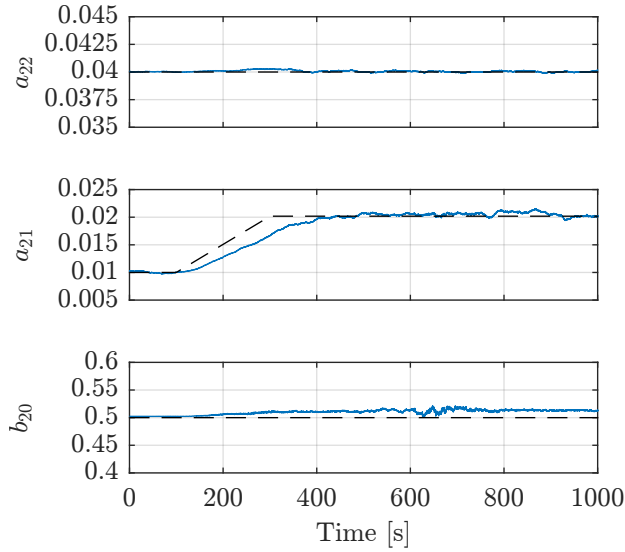


Figure 8.5: Parameter estimates (—) and true parameters (---) related to the second submodel with decreasing damping coefficient ζ_2 , which corresponds to an increasing a_{21} .

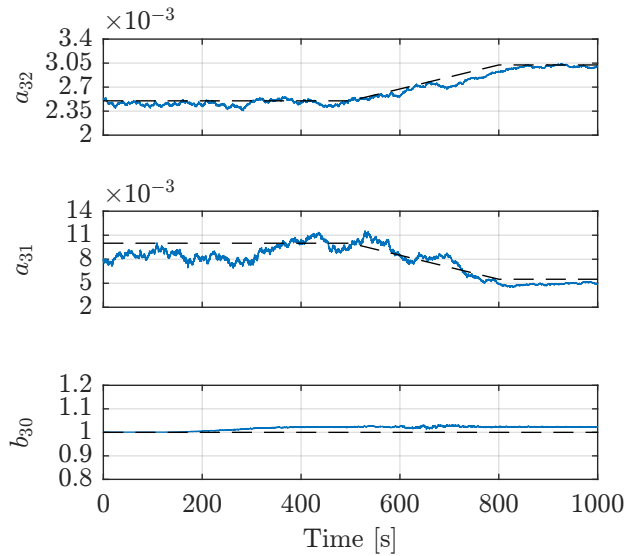


Figure 8.6: Parameter estimates (—) and true parameters (---) related to the third submodel with decreasing natural frequency ω_3 , which corresponds to an increasing a_{32} and decreasing a_{31} .

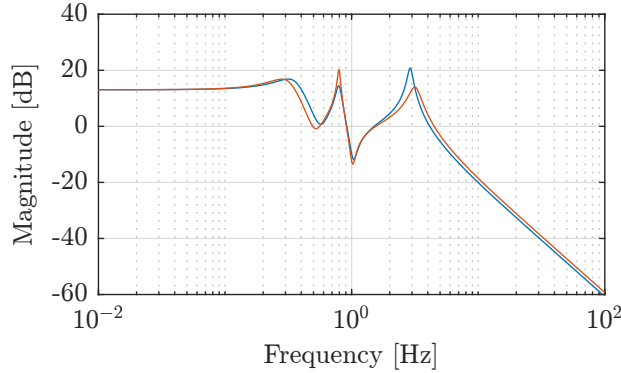


Figure 8.7: Plant estimate at $t = 50$ s (—) and at $t = 950$ s (—). Clearly, all resonances changed. Note that the natural frequency of the first mode decreased and the damping of the second mode decreased. A combination of the two occurred at the third mode.

8.6 Experimental validation on an overactuated and oversensed flexible beam setup

In this section, the method is tested on an experimental setup. To this end, consider the setup shown in Figure 8.8. The system consists of a thin flexible steel beam of $500 \times 20 \times 2$ mm. It is equipped with five contactless fiberoptic sensors and three voice-coil actuators and is suspended by wire flexures, leaving one rotational and one translational direction unconstrained. The system is operating at a sampling frequency of 4096 Hz. The second and fourth sensor are not used for the conducted experiments.

The main aim is to estimate the SISO system G^* between the input u , exciting the system at the outer ends of the beam equally, see Figures 8.8 and 8.9, and the output y which is the average deflection of the outer ends. This average deflection is controlled by a feedback controller and the setpoint is a square wave with a frequency of 2 Hz. By means of a second control loop, the internal stiffness of the beam is artificially manipulated, see [39, 43] for details.

Lightly-damped systems such as the one under study are well-suited to be described in a modal representation, similar to Example 8.3, i.e.,

$$G^*(p) = \sum_{i=1}^K \frac{b_{i,0}^*}{p^2/\omega_i^2 + 2(\zeta_i/\omega_i)p + 1},$$

where ζ_i and ω_i represent the damping coefficients and natural frequencies of the flexible modes respectively, see [91]. The flexible beam fits the modal description, and the system is approximated with $K = 2$ modes. The first mode is the

suspension

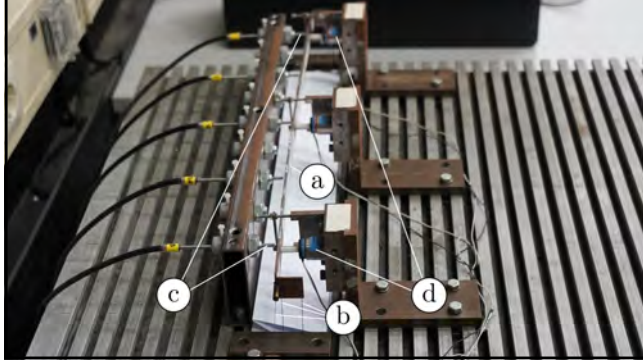


Figure 8.8: Prototype experimental flexible beam setup. The moving part is indicated by (a) and is suspended by wire flexures (b). The deflection is measured with five contactless fiber optic sensors, of which two are used (c) and the setup is actuated with three current-driven voice coils of which the outer two are used (d).

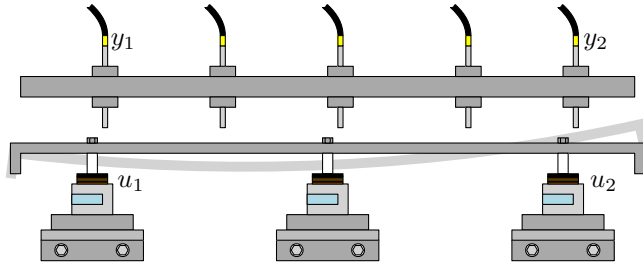


Figure 8.9: Schematic illustration of the experimental setup. The three actuators are visible on the bottom part of the image and the five sensors at the top. During operation, the beam translates, rotates and exhibits internal flexible behavior. The inputs and outputs are transformed as $u_1 = u_2 = \frac{1}{2}u$ and $y = \frac{1}{2}(y_1 + y_2)$, respectively.

mode of the system and the second mode is the first internal flexible mode which is artificially manipulated. Figure 8.12 shows the model of the system in the frequency domain, including the neglected higher order modes, and the model at t_0 after initializing the system. Two distinct forgetting factors are used tailored to the distinct modes, namely $\lambda_1 = 0.9999$ and $\lambda_2 = 0.999$, and the approximate covariance matrices are initialized at $\bar{P}_1(t_0) = \bar{P}_2(t_0) = \text{diag}(10^{-11}, 10^{-6}, 10^{-1})$.

An experiment is conducted where the internal flexible mode is actively manipulated from $t \in [40, 100]$ s. Algorithm 2 is deployed to estimate the time-

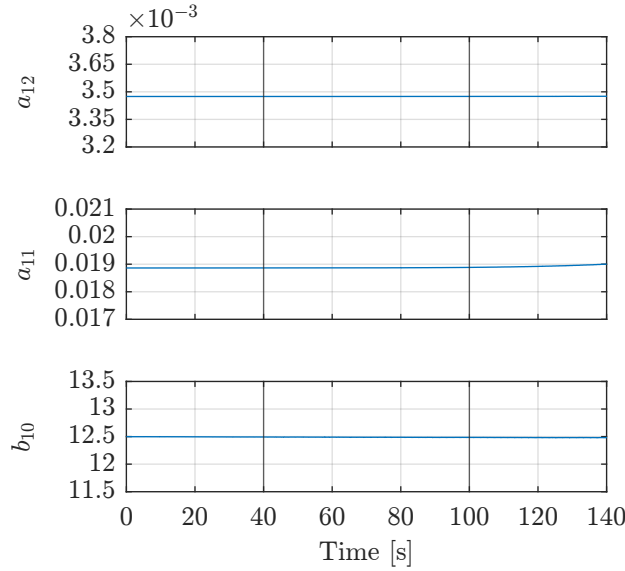


Figure 8.10: Parameter estimate related to the first submodel. This first mode relates to the suspension of the flexible beam setup which remains unaltered throughout the experiment explaining the graphs.

varying behavior, where no SRIVC iterations are performed and only a single coordinate descent iteration is performed at every incoming sample, i.e., $s = 0$ and $M_l = 1$. Since the system is operating in closed loop, the instrument is chosen as (8.33). The obtained estimates are depicted for each of the two submodels in Figures 8.10 and 8.11, and the estimated model at $t = 0$ s and $t = 135$ s is depicted in Figure 8.12. From the figures it is concluded that the shifting resonance mode is effectively observed, and the first mode does not exhibit parameter variations. The small fluctuations in the estimate of the second mode are at the frequency of the setpoint.

8.7 Conclusions

This chapter addresses a recursive method to identify additive systems for both open and closed-loop setups. The proposed algorithms, based on a block-coordinate descent with refined instrumental variables, are capable to track time-varying parameters of continuous-time systems in real-time. This allows tracking of more parsimonious and physically-relevant model representations, particularly for mechanical systems. Practical aspects are provided and finally the effectiveness of the proposed approach is illustrated through numerical vali-

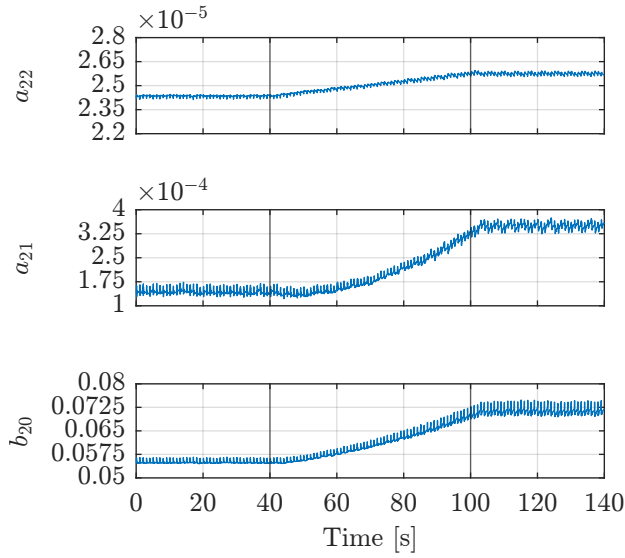


Figure 8.11: Parameter estimate related to the second submodel. This mode is artificially manipulated between $t = 40$ s and $t = 100$ s, which explains the changing parameters.

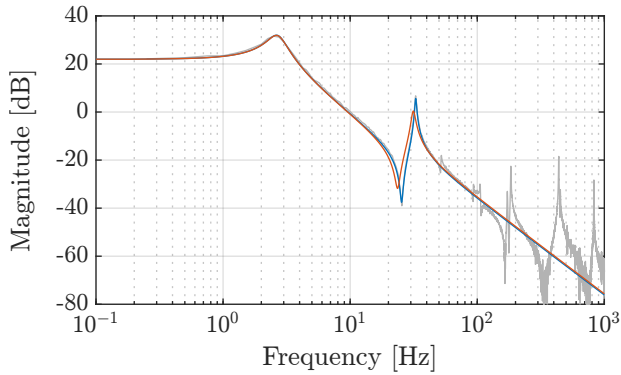


Figure 8.12: Frequency response function of the nonmanipulated system (—), the plant estimate at $t = t_{N_1} = 0$ s (—), and at the plant estimate at $t = 135$ s (—).

dation and application on a real-time setup.

Appendices

8.A Derivation of the refined instrumental variable iterations

In this appendix, it is shown that a stationary point of the following optimization problem is obtained from the refined instrumental variable method. Recall the optimization problem

$$\theta_i^{l+1}(t_N) = \arg \min_{\theta_i \in \Omega_i} \sum_{k=1}^N \alpha_{N-k} \left[\tilde{y}_i(t_k) - G_i(p, t, \theta_i) u(t_k) \right]^2, \quad (8.36)$$

with

$$\tilde{y}_i(t_k) := y(t_k) - \sum_{j=1}^{i-1} G_j(p, t, \theta_j^{l+1}) u(t_k) - \sum_{j=i+1}^K G_j(p, t, \theta_j^l) u(t_k).$$

and define

$$\varepsilon_i(t_k, \theta_i) = \tilde{y}_i(t_k) - G_i(p, t, \theta_i) u(t_k).$$

By taking Remark 8.17 into consideration, this residual is well approximated as

$$\varepsilon_i(t_k, \theta_i) = \frac{1}{A_i(p, \theta_i)} (A_i(p, \theta_i) \tilde{y}_i(t_k) - B_i(p, \theta_i) u(t_k)),$$

and subsequently $A_i(p, \theta_i)$ and $B_i(p, \theta_i)$ gives

$$\begin{aligned} \varepsilon_i(t_k, \theta_i) = \frac{1}{A_i(p, \theta_i)} & \left(a_{i,n_i}(t_k) p^{n_i} \tilde{y}_i(t_k) + \dots \right. \\ & + a_{i,1}(t_k) p \tilde{y}_i(t_k) + \tilde{y}_i(t_k) - b_{i,m_i}(t_k) p^{m_i} u(t_k) - \dots \\ & \left. - b_{i,1}(t_k) p u(t_k) - b_{i,0}(t_k) u(t_k) \right). \end{aligned}$$

The filtered residual output, filtered regressor vector, and parameter vector are defined as

$$\begin{aligned}\tilde{y}_{i,f}(t_k, \theta_i) &= \frac{1}{A_i(p, \theta_i)} \tilde{y}_i(t_k), \\ \varphi_{i,f}(t_k) &= \frac{1}{A_i(p, \theta_i)} \begin{bmatrix} -p\tilde{y}_i(t_k) & \dots & -p^{n_i}\tilde{y}_i(t_k) & u(t_k) & pu(t_k) & \dots & p^{m_i}u(t_k) \end{bmatrix}^\top, \\ \theta_i(t_k) &= \begin{bmatrix} a_{i,1}(t_k) & \dots & a_{i,n_i}(t_k) & b_{i,0}(t_k) & b_{i,1}(t_k) & \dots & b_{i,m_i}(t_k) \end{bmatrix}^\top,\end{aligned}$$

which allows to write ε_i as

$$\varepsilon_i(t_k, \theta_i) = \tilde{y}_{i,f}(t_k, \theta_i) - \varphi_{i,f}^\top(t_k, \theta_i)\theta_i(t_k), \quad (8.37)$$

which is a pseudolinear regression problem in θ_i that admits the refined instrumental variable [88, 277] iterations. To this end, the parameter estimate of the previous SRIVC iterate s is taken in the denominator polynomial A_i which is used as prefilter for $\tilde{y}_{i,f}$ and $\varphi_{i,f}$. Then, substitution in the first order optimality condition of (8.36), i.e.,

$$\sum_{k=1}^N \alpha_{N-k} \frac{\partial \varepsilon_i(t_k, \theta_i)}{\partial \theta_i} \varepsilon_i(t_k, \theta_i) = 0,$$

with the instrument vector $\hat{\varphi}_{i,f}(t_k, \theta_i)$ gives

$$\sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_i) (\tilde{y}_{i,f}(t_k, \theta_i) - \varphi_{i,f}^\top(t_k, \theta_i)\theta_i(t_k)) = 0.$$

Hence, the iterations for a fixed block descent iteration $l+1$ are of the form

$$\begin{aligned}\theta_{i,s+1}^{l+1}(t_N) &= \left[\sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_k, \theta_{i,s}^{l+1}) \right]^{-1} \\ &\quad \times \left[\sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \tilde{y}_{i,f}(t_k, \theta_{i,s}^{l+1}) \right],\end{aligned} \quad (8.38)$$

with $s \geq 1$.

8.B Recursive computation of the SRIVC iterations

In order to reformulate (8.25) into its recursive form, define

$$R_{i,s}^{l+1}(t_N) = \sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_k, \theta_{i,s}^{l+1}), \quad (8.39)$$

and

$$f_{i,s}^{l+1}(t_N) = \sum_{k=1}^N \alpha_{N-k} \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \tilde{y}_{i,f}(t_k, \theta_{i,s}^{l+1}). \quad (8.40)$$

Using that $\alpha_k = \lambda^k$ and that α satisfies the multiplicative property $\alpha_{k+1} = \lambda \alpha_k$, with $\alpha_0 = 1$,

$$R_{i,s}^{l+1}(t_k) = \lambda \bar{R}_i(t_{k-1}) + \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_k, \theta_{i,s}^{l+1}), \quad (8.41)$$

and

$$f_{i,s}^{l+1}(t_k) = \lambda \bar{f}_i(t_{k-1}) + \hat{\varphi}_{i,f}(t_k, \theta_{i,s}^{l+1}) \tilde{y}_{i,f}(t_k, \theta_{i,s}^{l+1}), \quad (8.42)$$

where $\bar{R}_i(t_{k-1})$ and $\bar{f}_i(t_{k-1})$ denote the final value of the iterations at the previous time step t_{k-1} (i.e., after iterating over the modes through i and l , and after the iterations in s). Using that $\theta_{i,s+1}^{l+1}(t_N) = (R_{i,s}^{l+1}(t_N))^{-1} f_{i,s}^{l+1}(t_N)$, and $\bar{f}_i(t_{k-1}) = \bar{R}_i(t_{k-1}) \bar{\theta}_i(t_{k-1})$,

$$\theta_{i,s+1}^{l+1}(t_N) = (R_{i,s}^{l+1}(t_N))^{-1} (\lambda \bar{R}_i(t_{N-1}) \bar{\theta}_i(t_{N-1}) + \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \tilde{y}_{i,f}(t_N, \theta_{i,s}^{l+1})),$$

which after substitution of $\bar{R}_i(t_{k-1})$ from (8.41) is written as

$$\begin{aligned} \theta_{i,s+1}^{l+1}(t_N) &= (R_{i,s}^{l+1}(t_N))^{-1} (\hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \tilde{y}_{i,f}(t_N, \theta_{i,s}^{l+1}) \\ &\quad + (R_{i,s}^{l+1}(t_N) - \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1})) \bar{\theta}_i(t_{N-1})), \end{aligned}$$

and thus

$$\begin{aligned} \theta_{i,s+1}^{l+1}(t_N) &= \bar{\theta}_i(t_{N-1}) + (R_{i,s}^{l+1}(t_N))^{-1} \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \\ &\quad \times (\tilde{y}_{i,f}(t_N, \theta_{i,s}^{l+1}) - \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \bar{\theta}_i(t_{N-1})). \end{aligned} \quad (8.43)$$

In order to avoid inverting $R_{i,s}^{l+1}(t_N)$ at each time step, introduce $P_{i,s}^{l+1}(t_N) = (R_{i,s}^{l+1}(t_N))^{-1}$. The Sherman–Morrison formula [133, Chap. 0.7.4], a generalization of the matrix inversion lemma, states that $(A + uv^\top)^{-1} = A^{-1} - \frac{A^{-1}uv^\top A^{-1}}{1 + v^\top A^{-1}u}$, provided the inverses exist, with A invertible and square, and column vectors u and v . Take $A = \lambda \bar{R}_i(t_{N-1})$, $u = \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1})$, and $v = \varphi_{i,f}(t_N, \theta_{i,s}^{l+1})$. Then,

$$\begin{aligned} (R_{i,s}^{l+1}(t_N))^{-1} &= \frac{1}{\lambda} (\bar{R}_i(t_{N-1}))^{-1} \\ &\quad - \frac{\frac{1}{\lambda} (\bar{R}_i(t_{N-1}))^{-1} \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \frac{1}{\lambda} (\bar{R}_i(t_{N-1}))^{-1}}{1 + \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \frac{1}{\lambda} (\bar{R}_i(t_{N-1}))^{-1} \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1})}, \end{aligned}$$

and thus with $\bar{P}_i(t_{N-1}) = (\bar{R}_i(t_{N-1}))^{-1}$,

$$P_{i,s}^{l+1}(t_N) = \frac{1}{\lambda} \bar{P}_i(t_{N-1}) - \frac{1}{\lambda} \frac{\bar{P}_i(t_{N-1}) \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \bar{P}_i(t_{N-1})}{\lambda + \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \bar{P}_i(t_{N-1}) \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1})}. \quad (8.44)$$

Define $K_{i,s+1}^{l+1}(t_N) = P_{i,s}^{l+1}(t_N) \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1})$, or equivalently,

$$K_{i,s+1}^{l+1}(t_N) = \bar{P}_i(t_{N-1}) \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}) \times (\lambda + \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \bar{P}_i(t_{N-1}) \hat{\varphi}_{i,f}(t_N, \theta_{i,s}^{l+1}))^{-1}. \quad (8.45)$$

This expression allows us to write (8.44) as

$$P_{i,s}^{l+1}(t_N) = \frac{1}{\lambda} (\bar{P}_i(t_{N-1}) - K_{i,s+1}^{l+1}(t_N) \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \bar{P}_i(t_{N-1})),$$

and (8.43) as

$$\theta_{i,s+1}^{l+1}(t_N) = \bar{\theta}_i(t_{N-1}) + K_{i,s+1}^{l+1}(t_N) (\tilde{y}_{i,f}(t_N, \theta_{i,s}^{l+1}) - \varphi_{i,f}^\top(t_N, \theta_{i,s}^{l+1}) \bar{\theta}_i(t_{N-1})).$$

Hence, (8.45), (8.44), and (8.43), form a recursive equivalent of (8.38).

8.C Proof of theorem 8.14

In this appendix, the proof of Theorem 8.14 is provided.

Proof. As the number of SRIVC iterations s tends to infinity,

$$\begin{aligned} \theta_{i,\infty}^{l+1} &= \frac{1}{N} \left[\sum_{k=1}^N \hat{\varphi}_{i,f}(t_k, \theta_{i,\infty}^{l+1}) \varphi_{i,f}^\top(t_k, \theta_{i,\infty}^{l+1}) \right]^{-1} \\ &\times \frac{1}{N} \left[\sum_{k=1}^N \hat{\varphi}_{i,f}(t_k, \theta_{i,\infty}^{l+1}) \tilde{y}_{i,f}(t_k, \theta_{i,\infty}^{l+1}) \right]. \end{aligned} \quad (8.46)$$

That is,

$$\frac{1}{N} \sum_{k=1}^N \hat{\varphi}_{i,f}(t_k, \theta_{i,\infty}^{l+1}) [\tilde{y}_{i,f}(t_k, \theta_{i,\infty}^{l+1}) - \varphi_{i,f}^\top(t_k, \theta_{i,\infty}^{l+1}) \theta_{i,\infty}^{l+1}] = 0, \quad (8.47)$$

where

$$\begin{aligned} & \tilde{y}_{i,f}(t_k, \theta_{i,\infty}^{l+1}) - \varphi_{i,f}^\top(t_k, \theta_{i,\infty}^{l+1}) \theta_{i,\infty}^{l+1} \\ &= \frac{1}{A_i(t_k, \theta_{i,\infty}^{l+1})} \tilde{y}_i(t_k) + \frac{p a_{i,\infty,1}^{l+1} + \dots + p^{n_i} a_{i,\infty,n_i}^{l+1}}{A_i(t_k, \theta_{i,\infty}^{l+1})} \tilde{y}_i(t_k) \\ & \quad - \frac{b_{i,\infty,0}^{l+1} + p b_{i,\infty,1}^{l+1} + \dots + p^{m_i} b_{i,\infty,m_i}^{l+1}}{A_i(t_k, \theta_{i,\infty}^{l+1})} u(t_k) \end{aligned} \quad (8.48)$$

$$= \tilde{y}_i(t_k) - \frac{B_i(t_k, \theta_{i,\infty}^{l+1})}{A_i(t_k, \theta_{i,\infty}^{l+1})} u(t_k) \quad (8.49)$$

$$\begin{aligned} &= y(t_k) - \sum_{j=1}^{i-1} G_j(p, t_k, \theta_j^{l+1}) u(t_k) \\ & \quad - \sum_{j=i+1}^K G_j(p, t_k, \theta_j^l) u(t_k) - \frac{B_i(t_k, \theta_{i,\infty}^{l+1})}{A_i(t_k, \theta_{i,\infty}^{l+1})} u(t_k). \end{aligned} \quad (8.50)$$

To simplify the notation, in the subsequent analysis, the t_k dependency on the transfer functions is omitted. Denoting the parameter vector $\theta_{i,\infty}^\infty$ as the limit of $\theta_{i,\infty}^l$ when the number of descent iterations tends to infinity, then, as $l \rightarrow \infty$,

$$\tilde{y}_{i,f}(t_k, \theta_{i,\infty}^\infty) - \varphi_{i,f}^\top(t_k, \theta_{i,\infty}^\infty) \theta_{i,\infty}^\infty = y(t_k) - \sum_{j=1}^K G_j(p, \theta_{j,\infty}^\infty) u(t_k). \quad (8.51)$$

On the other hand, note that

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N \hat{\varphi}_{i,f}(t_k, \theta_{i,\infty}^\infty) v(t_k) = \bar{\mathbb{E}} \{ \hat{\varphi}_{i,f}(t_k, \theta_{i,\infty}^\infty) v(t_k) \} = 0, \quad (8.52)$$

where the first equality is due to the ergodicity property of the input and noise signals (see Theorem 2B.1 of [31] for more details), and the second equality holds since v and $\hat{\varphi}_{i,f}$ are uncorrelated due to Assumption 8.12. So, for $i = 1, \dots, K$, combining (8.47), (8.51), (8.13) and (8.52) leads to

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{j=1}^K \sum_{k=1}^N \hat{\varphi}_{i,f}(t_k, \theta_{i,\infty}^\infty) [G_j^*(p) - G_j(p, \theta_{j,\infty}^\infty)] u(t_k) = 0. \quad (8.53)$$

Now,

$$\begin{aligned}
 G_j^*(p) - G_j(p, \theta_{j,\infty}^\infty) &= \frac{B_j^*(p)}{A_j^*(p)} - \frac{B_j(p, \theta_{j,\infty}^\infty)}{A_j(p, \theta_{j,\infty}^\infty)} \\
 &= \frac{B_j^*(p)A_j(p, \theta_{j,\infty}^\infty) - A_j^*(p)B_j(p, \theta_{j,\infty}^\infty)}{A_j^*(p)A_j(p, \theta_{j,\infty}^\infty)} \\
 &= \frac{[1 \ p \ \dots \ p^{n_j+m_j}]}{A_j^*(p)A_j(p, \theta_{j,\infty}^\infty)} h_j,
 \end{aligned} \tag{8.54}$$

where $h_j \in \mathbb{R}^{n_j+m_j+1}$ is the vector that contains the coefficients of $B_j^*(p)A_j(p, \theta_{j,\infty}^\infty) - A_j^*(p)B_j(p, \theta_{j,\infty}^\infty)$ in increasing order of degree. With regards to the instrument vector, the following decomposition holds (see, e.g., Eq. (16) of [191])

$$\hat{\varphi}_{i,t}(t_k, \theta_{i,\infty}^\infty) = S(-B_i, A_i) \frac{1}{A_i^2(p, \theta_{i,\infty}^\infty)} U_i(t_k), \tag{8.55}$$

where $U_i(t_k) = [1 \ p \ \dots \ p^{n_i+m_i}]^\top u(t_k)$, and where $S(-B_i, A_i)$ is a Sylvester matrix formed by the coefficients of the polynomials $-B_i(p, \theta_{i,\infty}^\infty)$ and $A_i(p, \theta_{i,\infty}^\infty)$, similar to Eq. (13) of [191]. Thus, (8.53) is equivalent to

$$\begin{aligned}
 &\sum_{j=1}^K \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N S(-B_i, A_i) \frac{1}{A_i^2(t_k, \theta_{i,\infty}^\infty)} U_i(t_k) \\
 &\quad \times \frac{1}{A_j^*(p)A_j(p, \theta_{j,\infty}^\infty)} U_j^\top(t_k) h_j = 0,
 \end{aligned} \tag{8.56}$$

for $i = 1, \dots, K$. After applying Lemma 8.19 in Section 8.C.1 and the ergodicity result in Theorem 2B.1 of [31], which hold since the system and model denominators describe uniformly stable transfer functions and the input is assumed quasi-stationary, the following is obtained.

$$S(-\bar{B}_i, \bar{A}_i) \sum_{j=1}^K \mathbb{E} \left\{ \frac{1}{A_i^2(t_k, \theta_{i,\infty}^\infty)} U_i(t_k) \frac{1}{A_j^*(p)A_j(p, \theta_{j,\infty}^\infty)} U_j^\top(t_k) \right\} \bar{h}_j = 0, \tag{8.57}$$

for $i = 1, \dots, K$, where $(\bar{A}_i, \bar{B}_i) = \lim_{N \rightarrow \infty} (A_i(p, \theta_{i,\infty}^\infty), B_i(p, \theta_{i,\infty}^\infty))$, and $\bar{h}_j = \lim_{N \rightarrow \infty} h_j$. The K conditions in (8.57) are jointly expressed as $\mathcal{S} \Phi_u \eta = 0$, where $\mathcal{S}$ is a block-diagonal matrix whose block-diagonal is comprised by $\{S(-\bar{B}_i, \bar{A}_i)\}_{i=1}^K$, while $\eta = [\bar{h}_1^\top, \dots, \bar{h}_K^\top]^\top$, and Φ_u has block entries given by

$$\Phi_u^{i,j} = \mathbb{E} \left\{ \frac{1}{A_i^2(t_k, \theta_{j,\infty}^\infty)} U_i(t_k) \frac{1}{A_j^*(p)A_j(p, \theta_{j,\infty}^\infty)} U_j(t_k) \right\}.$$

Assumption 8.11 regarding the coprimeness of $\bar{A}_i$ and $\bar{B}_i$ implies that $\mathcal{S}$ is nonsingular, as shown in Lemma A3.1 of [234]. Moreover, Lemma 8.20 in Section 8.C.1

applied over each entry of Φ_u indicates that the non-singularity of Φ_u can be studied directly from the cross covariance of the inputs filtered by the LTI systems obtained at the limit as $N \rightarrow \infty$. Thus, following the same reasoning as in Lemma 3 of [106], Assumptions 8.9 to 8.12 imply the generic nonsingularity of Φ_u with respect to the converging system and model denominator parameters. Consequently, $\eta = 0$, i.e., $\lim_{N \rightarrow \infty} h_i = 0$ for all $i = 1, \dots, K$, as desired. $\square$

8.C.1 Technical Lemmas

Lemma 8.19. *If the matrix $\mathcal{O}_k$ has a bounded norm for all k with probability 1, then, with probability 1,*

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N S_k \mathcal{O}_k h_k = \bar{S} \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{k=1}^N \mathcal{O}_k \bar{h}, \quad (8.58)$$

where S_k and h_k are matrices and vectors of appropriate dimensions, such that $\lim_{k \rightarrow \infty} S_k = \bar{S}$ and $\lim_{k \rightarrow \infty} h_k = \bar{h}$.

Proof. Note that the following inequalities hold:

$$\frac{1}{N} \left\| \sum_{k=1}^N (S_k - \bar{S}) \mathcal{O}_k h_k \right\| \leq \frac{1}{N} \sum_{k=1}^N \|S_k - \bar{S}\| \|\mathcal{O}_k h_k\| \leq \frac{M_1}{N} \sum_{k=1}^N \|S_k - \bar{S}\|, \quad (8.59)$$

where $M_1 = \sup_k \|\mathcal{O}_k h_k\| < \infty$ with probability 1, since $\mathcal{O}_k$ and h_k are bounded. Since $\|S_k - \bar{S}\| \rightarrow 0$ as $k \rightarrow \infty$, then the Cesàro mean in (8.59) also goes to zero due to Theorem 4.2.3 of [51]. By the same reasoning, it follows that

$$\frac{1}{N} \left\| \sum_{k=1}^N \bar{S} \mathcal{O}_k (h_k - \bar{h}) \right\| \leq \frac{1}{N} \sum_{k=1}^N \|\bar{S} \mathcal{O}_k\| \|h_k - \bar{h}\| \leq \frac{M_2}{N} \sum_{k=1}^N \|h_k - \bar{h}\| \xrightarrow{N \rightarrow \infty} 0.$$

Thus, the result in (8.58) follows from decomposing $S_k \mathcal{O}_k h_k$ as

$$S_k \mathcal{O}_k h_k = (S_k - \bar{S}) \mathcal{O}_k h_k + \bar{S} \mathcal{O}_k (h_k - \bar{h}) + \bar{S} \mathcal{O}_k \bar{h},$$

summing over k , dividing over N , and computing the limit as $N \rightarrow \infty$. $\square$

Lemma 8.20. *Let*

$$w(t) = \sum_{k=0}^{\infty} \alpha_t(k) e(t-k), \quad v(t) = \sum_{k=0}^{\infty} \beta_t(k) e(t-k), \quad (8.60)$$

where the families of filters $\alpha_t(k), \beta_t(k)$, $t = 1, 2, \dots$ are uniformly stable, and $\{e(t)\}$ is a quasi-stationary signal. In addition, let $\bar{w}(t)$ and $\bar{v}(t)$ be defined as in (8.60) but for the LTI filters $\bar{\alpha}(k)$ and $\bar{\beta}(k)$ respectively, where $\alpha_t(k) \rightarrow \bar{\alpha}(k)$ and $\beta_t(k) \rightarrow \bar{\beta}(k)$ for every k as $t \rightarrow \infty$. Then,

$$|\mathbb{E}\{w(t)v(t)\} - \mathbb{E}\{\bar{w}(t)\bar{v}(t)\}| \rightarrow 0 \quad \text{as } N \rightarrow \infty. \quad (8.61)$$

Proof. The following inequalities hold:

$$\begin{aligned} & |\bar{\mathbb{E}}\{w(t)v(t)\} - \bar{\mathbb{E}}\{\bar{w}(t)\bar{v}(t)\}| \\ &= \left| \frac{1}{N} \sum_{t=1}^N \sum_{k=0}^{\infty} \sum_{\ell=0}^{\infty} \mathbb{E}\{e(t-k)e(t-\ell)\}(\alpha_t(k)\beta_t(\ell) - \bar{\alpha}(k)\bar{\beta}(\ell)) \right| \end{aligned} \quad (8.62)$$

$$\leq \frac{C}{N} \sum_{t=1}^N \sum_{k=0}^{\infty} \sum_{\ell=0}^{\infty} |(\alpha_t(k)(\beta_t(\ell) - \bar{\beta}(\ell)) + \bar{\beta}(\ell)(\alpha_t(k) - \bar{\alpha}(k)))| \quad (8.63)$$

$$\begin{aligned} &\leq \frac{C}{N} \sup_{t \in \mathbb{N}} \sum_{k=0}^{\infty} |\alpha_t(k)| \sum_{t=1}^N \sum_{\ell=0}^{\infty} |\beta_t(\ell) - \bar{\beta}(\ell)| \\ &+ \frac{C}{N} \sum_{\ell=0}^{\infty} |\bar{\beta}(\ell)| \sum_{t=1}^N \sum_{k=0}^{\infty} |\alpha_t(k) - \bar{\alpha}(k)|, \end{aligned} \quad (8.64)$$

where (8.62) holds by definition, (8.63) uses the quasi-stationary property $|\mathbb{E}\{e(t-k)e(t-\ell)\}| \leq C$ for any indices t, k and ℓ , and (8.64) is due to the triangle inequality. Since $\alpha_t(k)$ and $\beta_t(\ell)$ describe uniformly stable families of filters, the series $\sup_{t \in \mathbb{N}} \sum_{k=0}^{\infty} |\alpha_t(k)|$ and $\sum_{k=0}^{\infty} |\bar{\beta}(k)|$ converge. Thus, to prove (8.61) it is required to show that

$$\frac{1}{N} \sum_{t=1}^N \sum_{k=0}^{\infty} |\alpha_t(k) - \bar{\alpha}(k)| \rightarrow 0 \quad \text{as } N \rightarrow \infty, \quad (8.65)$$

and similarly with $\beta_t(k)$. Due to the pointwise convergence, $\lim_{t \rightarrow \infty} |\alpha_t(k) - \bar{\alpha}(k)| = 0$, and due to the uniform stability of $\alpha_t(k)$,

$$\sum_{k=0}^{\infty} |\alpha_t(k) - \bar{\alpha}(k)| \leq 2 \sum_{k=0}^{\infty} \sup_{t \in \mathbb{N}} |\alpha_t(k)| < \infty.$$

Thus, by the Weierstrass M-test [20, Theorem A28],

$$\lim_{t \rightarrow \infty} \sum_{k=0}^{\infty} |\alpha_t(k) - \bar{\alpha}(k)| = \sum_{k=0}^{\infty} \lim_{t \rightarrow \infty} |\alpha_t(k) - \bar{\alpha}(k)| = 0.$$

Hence, (8.65) can be interpreted as a Cesàro mean with sequence in t given by $\sum_{k=0}^{\infty} |\alpha_t(k) - \bar{\alpha}(k)|$, which goes to zero as t tends to infinity. Thus, by Theorem 4.2.3 of [51], (8.65) holds. The same derivation holds for $\beta_t(\ell)$, which implies that the right hand side of (8.64) tends to zero as N tends to infinity, thus proving (8.61). $\square$